Defining Effective Rosseland Mean for Expansion Opacity

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Abstract

The standard Rosseland mean does not account for the Doppler shift of photons through spectral lines during expansion and can substantially misestimate the opacity of line-rich ejecta. We define an effective Rosseland mean opacity, $\kappa_R^{\rm exp}$, for radiative transfer in expanding media where bound-bound transitions dominate the opacity. For each frequency, the method calculates the distance an outward travelling photon covers before reaching optical depth unity, and assigns an effective expansion opacity $\kappa_{\rm exp} = 1/(\rho\Delta x)$. Single-group Monte Carlo radiative transfer simulations using this opacity reproduce the trends of full frequency-group calculations for Type Ia supernova and kilonova ejecta, with peak luminosities differing by 10–30%, while reducing the computational time by more than an order of magnitude. For the kilonova we use a composition of eight representative elements derived from the solar r-process abundance pattern, and show that the single-group calculation follows the multigroup light curve over four decades in lanthanide mass fraction.

Keywords:
opacity, Rosseland mean, expansion opacity, kilonova, supernovae

1. Introduction

The interaction between radiation and matter is primarily determined by the material properties such as composition, density, and temperature, as well as the spectral characteristics of the radiation. In expanding plasmas, due to the Doppler effect, as photons travel through the plasma, their frequency changes. This effect is mainly important for bound-bound transitions, because their narrow line structure in frequency causes photons to interact with different lines as they travel through the plasma. This effect is known as expansion opacity, and it is important for many astrophysical applications, such as Type Ia supernovae, kilonovae and stellar winds.

For non-moving plasmas, the radiation flux can be approximated well using the Rosseland mean opacity, which is a weighted average of the opacity over all frequencies. However, for expanding plasmas with significant bound-bound opacity, the Rosseland mean opacity is not a good approximation, since the width of the bb lines is negligible and photons interact with them only due to their frequency change as they travel through the plasma. Therefore the average distance a photon travels cannot be derived from $\kappa(\nu)$ alone because it inherently depends on the photon's direction of movement and the plasma's velocity gradient. To avoid this problem and to allow calculation of transport that can inherently depend on the photon's direction of movement and velocity gradient, radiation transport in systems with expansion opacity is usually calculated using Monte Carlo simulations, which track individual photons and their frequency evolution as they propagate through the expanding medium.

Calculating the Rosseland mean opacity has several motivations: a single opacity value (1) is required for analytic models and closed-form solutions, which cannot treat full frequency-dependent opacity; (2) gives direct physical intuition for how temperature and composition affect the opacity and hence the light curve; and (3) enables faster and simpler simulations than full frequency-dependent calculations.

Previous attempts to address this problem have been proposed. Fontes et al. [1, 2] suggested using artificial line broadening by assigning a line width corresponding to unphysical high temperature of 20 MeV, which effectively smears the bound-bound lines over a finite frequency range. This approach preserves the integral of the opacity over frequency (area-preserving line profiles) and for NSM ejecta it produces frequency-dependent opacities that are one to two orders of magnitude greater than those obtained with expansion opacities in the Sobolev approximation without broadening. Fontes et al. [3] demonstrated that their smeared opacity tables, calculated using the Sobolev formalism, yield light curves for multigroup radiative transfer simulations that are similar to those obtained from full line-by-line Monte Carlo radiation transport (using the superNu code) for kilonovae (neutron star mergers). However, this approach still requires multigroup frequency-dependent opacity tables rather than a single gray (frequency-averaged) Rosseland mean opacity and uses an unphysical assumption that has been verified only for specific ejecta. Several works [4, 5, 6] have shown how opacity depends on composition, temperature, density, and time by comparing the Planck mean expansion opacity over those parameters. The Planck mean is nevertheless not the appropriate average for the opacity that controls photon escape from the ejecta. In a non-expanding medium this transport opacity is the Rosseland mean, which

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weights the low-opacity frequency windows through which photons preferentially diffuse. In an expanding medium the same transport character is needed, but the opacity must also include the line absorption encountered as the photon redshifts through the expanding flow.

In this paper we present the effective expansion opacity, $\kappa_{\rm exp}$, which accounts for bound-bound expansion opacity. The method follows an outward-moving photon in homologous expansion ($r = vt$) and calculates the distance over which it reaches optical depth $\tau \simeq 1$. Taking the Rosseland mean of $\kappa_{\rm exp}$ defines the effective expansion Rosseland mean, $\kappa_R^{\rm exp}(T, \rho, t)$. This gray-opacity method assumes that the photons dominating the diffusion are close to a blackbody spectrum. It is therefore useful for bolometric luminosities and characteristic temperatures, but not for detailed line spectra.

The structure of the paper is as follows: In Section 2 we describe how the expansion opacity is calculated, and define the effective expansion opacity. In Section 3, we show the variation of $\kappa_R^{\rm exp}$ with composition, temperature, density, and time and show how it scales differently from the Planck mean. In Section 4 we demonstrate the use of $\kappa_R^{\rm exp}$ in radiative transfer simulations and compare the results to the full frequency-group calculations. In Section 6 we summarize the results.

2. Expansion Opacity

The total opacity κ is the sum of contributions from different physical processes:

$$\kappa = \kappa_{\rm T} + \kappa_{\rm ff} + \kappa_{\rm bf} + \kappa_{\rm bb} \tag{1}$$

where $\kappa_{\rm T}$ is Thomson scattering, $\kappa_{\rm ff}$ is free-free opacity, $\kappa_{\rm bf}$ is bound-free opacity (which we neglect in this work), and $\kappa_{\rm bb}$ is bound-bound opacity from atomic line transitions.

The bb opacity $\kappa_{\rm bb}$ depends on the ionization and excitation populations of the material. We obtain the ionization fractions from the Saha equation and the level populations from the Boltzmann distribution, and use them to calculate the optical depth of each line.

The optical depth of the bound-bound line is then calculated using the Sobolev approximation [7].

$$\tau_{j_l,j_u} = \frac{\pi e^2}{m_e} \frac{f_{j_l,j_u}}{\nu_{j_l,j_u}} t \left(N_{m,i,j_l} - \frac{g_{j_l}}{g_{j_u}} N_{m,i,j_u} \right) \tag{2}$$

where m is the material index, i is the ionization state, j_l and j_u are the lower and upper energy levels respectively, f_{j_l,j_u} is the oscillator strength of the transition from the energy level j_l to the energy level j_u, ν_{j_l,j_u} is the frequency of the transition, N_{m,i,j_l} and N_{m,i,j_u} are the number densities of the lower and upper levels respectively, g_{j_l} and g_{j_u} are their degeneracies, and t is the time from explosion.

There are two values for the bound-bound opacity, saturated and unsaturated which are calculated as:

$$\kappa_{bb,saturated}(\nu) = \sum_l \frac{\nu_l}{\rho ct} \delta(\nu - \nu_l) \cdot (1 - \exp(-\tau_{j_l,j_u})) \tag{3}$$

$$\kappa_{bb,unsaturated}(\nu) = \sum_l \frac{\nu_l}{\rho ct} \delta(\nu - \nu_l) \cdot \tau_{j_l,j_u} \tag{4}$$

The saturated opacity gives the contribution of a narrow line to a larger finite frequency bin, by capping its effect once it becomes optically thick, since a single line has a maximal effect on the opacity once all photons in the line's frequency are absorbed. The unsaturated opacity gives the optical depth accumulated by a photon that crosses the full frequency bin.

We define the effective expansion opacity, $\kappa_{\rm exp}$, from the distance an outward travelling photon moves through the plasma before accumulating optical depth unity. The effective expansion Rosseland mean, $\kappa_R^{\rm exp}$, is then the Rosseland mean of $\kappa_{\rm exp}$. An outward travelling photon with frequency ν_0, will reach a new spectral line with frequency ν_l after a velocity shift of $\Delta v = -c \ln(\nu_0/\nu_l)$ which corresponds to a distance of $\Delta x = -ct \ln(\nu_0/\nu_l)$.

For that we define the cumulative frequency optical depth as:

$$\tau(\nu) = \int \kappa(\nu) \rho \, dx = ct \int \kappa(\nu) \rho \frac{d\nu}{\nu} \tag{5}$$

And for each frequency ν_i, given the frequency ν_j so that $\tau(\nu_j) - \tau(\nu_i) = 1$, we define the effective expansion opacity as:

$$\kappa_{\rm exp}(\nu_i) = \frac{1}{\rho \Delta x} = \frac{1}{\rho ct \ln(\nu_j/\nu_i)} \tag{6}$$

For finite size frequency bins this process is done by estimating where inside the stopping bin the photon reaches optical depth unity. This is needed because assigning the stopping point to the edge of a frequency bin would make the minimum travel distance depend on the chosen bin width, rather than on the line opacity inside the bin.

Numerically, after discretizing the frequency, the stopping bin is found with the unsaturated optical depth by finding the first bin j such that

$$\tau(\nu_{j-1}) - \tau(\nu_i) < 1, \qquad \tau(\nu_j) - \tau(\nu_i) > 1 \tag{7}$$

Unsaturated opacity is used for this step because it identifies where the photon reaches true optical depth unity; with saturated opacity alone, a strong line with $\tau \gg 1$ contributes only $1 - \exp(-\tau)$ and cannot set the stopping point by itself. We then estimate the sub-bin stopping frequency in bin j using the saturated opacity. If the saturated optical depth in this bin can supply the remaining optical depth to unity, we use

$$\Delta\nu_{\tau=1,i} = \nu_{j-1} - \nu_i + (\nu_j - \nu_{j-1}) \frac{1 - (\tau(\nu_{j-1}) - \tau(\nu_i))}{\Delta\tau_j^s},$$
$$\nu_{\tau=1,i} = \nu_i + \Delta\nu_{\tau=1,i}. \tag{8}$$

where $\Delta\tau_j^s$ is the saturated optical depth of bin j. Otherwise, if $\Delta\tau_j^s < 1 - (\tau(\nu_{j-1}) - \tau(\nu_i))$, we set $\Delta\nu_{\tau=1,i} = \nu_j - \nu_i$. The effective opacity is then calculated from $\Delta x_i = ct \ln(\nu_{\tau=1,i}/\nu_i)$. When a single bin dominates, we instead use

$$\kappa_{\rm exp}(\nu_i) = \frac{\sum_l (1 - \exp(-\tau(\nu_l)))}{\rho ct \ln(\nu_{i+1}/\nu_i)} + \kappa_{\rm T} + \kappa_{ff}(\nu_i) \tag{9}$$

This interpolation is approximate, but it is accurate when $\Delta\tau \ll 1$ or $\Delta\tau \gg 1$. For the intermediate case, $\Delta\tau \sim 1$, it introduces a small error which should mostly cancel on average.

Figure 1 shows the unsaturated opacity, saturated opacity, and effective expansion opacity of iron after 3 days, for density 1×10^{-12} g/cm^3 and temperature 0.5 eV. For species with $A < 28$ in Type Ia supernova compositions we use the atomic line data of Kurucz & Bell (1995; CD 23) [8]. For kilonova composition, where the relevant nuclei have $A > 28$, we use the NIST Atomic Spectra Database [9].

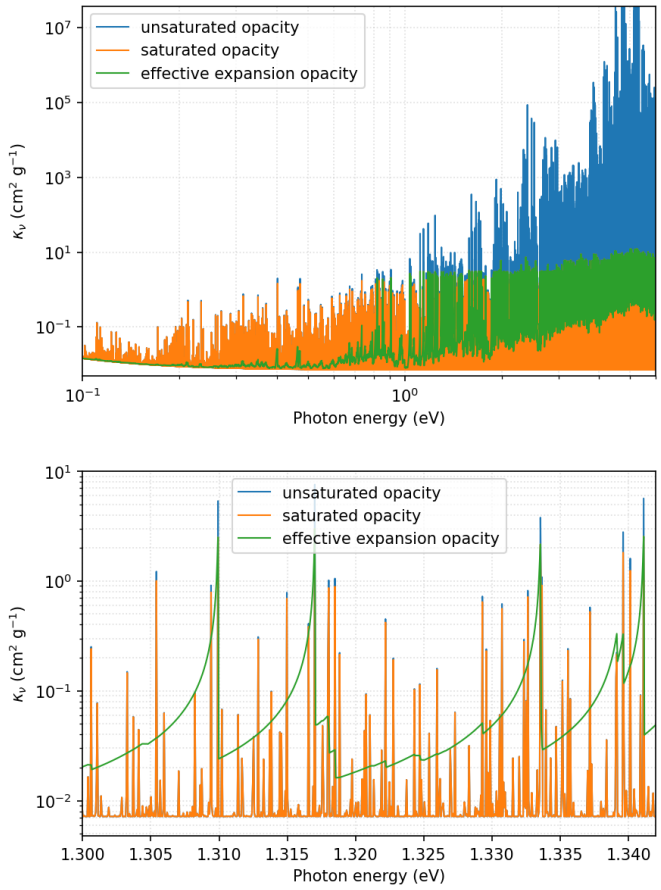


Figure 1: Unsaturated opacity, saturated opacity, and effective expansion opacity, for iron at $\rho = 1 \times 10^{-12}$ g/cm^3, $T = 0.5$ eV, $t = 3$ days. For large (top) and zoomed (bottom) energy ranges.

3. Rosseland Mean of Effective Expansion Opacity

The Planck mean opacity is an emission-weighted average, while the Rosseland mean opacity is a transport-weighted harmonic mean:

$$\kappa_P = \frac{\int \kappa_\nu B_\nu(T) d\nu}{\int B_\nu(T) d\nu}, \qquad \frac{1}{\kappa_R} = \frac{\int \frac{1}{\kappa_\nu} \frac{\partial B_\nu(T)}{\partial T} d\nu}{\int \frac{\partial B_\nu(T)}{\partial T} d\nu}. \tag{10}$$

where $B_\nu(T)$ is the Planck function. When the Rosseland average is applied to the usual frequency-dependent opacity κ_ν we denote the result by κ_R. When it is applied to the effective expansion opacity $\kappa_{\rm exp}$, we denote the result by $\kappa_R^{\rm exp}$.

The main problem with applying the Rosseland mean directly to the expansion opacity is that zero-width bound-bound lines cannot contribute to a frequency-integrated harmonic mean; numerically, this appears as a vanishing bound-bound contribution as the frequency bins are refined. Figure 2 shows the convergence of the Rosseland mean opacities as a function of the frequency resolution $\Delta\nu/\nu$ (for logarithmically spaced frequency bins) for iron (Fe, $Z = 26$) in top and erbium (Er, $Z = 68$) in bottom. Iron is an iron-group element with an open d-block, representative of the heavy material in Type Ia supernova ejecta. Erbium is a lanthanide with an open f-block, which is common in kilonova compositions and gives a much larger bound-bound opacity than iron. In Figure 2 we plot the ordinary Rosseland mean of the saturated expansion opacity, κ_R, and the effective expansion Rosseland mean, $\kappa_R^{\rm exp}$.

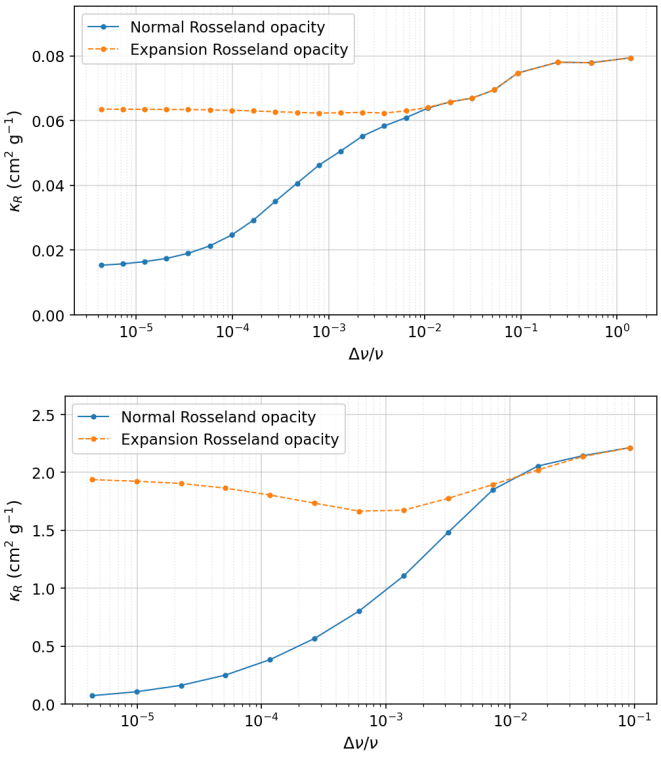


Figure 2: Convergence of the Rosseland mean opacities vs. the frequency resolution in $\Delta\nu/\nu$. Top: iron for Type Ia-like conditions, $\rho = 1 \times 10^{-13}$ g/cm^3, $T = 1$ eV, $t = 20$ days. Bottom: erbium (lanthanide, $Z = 68$); $\rho = 1 \times 10^{-12}$ g/cm^3, $T = 1$ eV, $t = 1$ day. In each panel the blue solid curve is the ordinary Rosseland mean κ_R of the saturated expansion opacity, and the orange dashed curve is the effective expansion Rosseland mean $\kappa_R^{\rm exp}$.

The results show that $\kappa_R^{\rm exp}$ is barely dependent on the frequency resolution and gives converged opacity even for low resolutions such as $\Delta\nu/\nu \gtrsim 4 \cdot 10^{-2}$ (which is equivalent to less than 200 bins for the full spectrum used). The reason the effective opacity gives correct results even for large bins is that the spectral lines are randomly distributed within each frequency bin. As long as the lines are randomly distributed, Eq. (9) provides a good approximation since it represents the typical distance a photon travels in a frequency bin. The fact that we observe good agreement between low-resolution and

high-resolution calculations for $\kappa_R^{\rm exp}$ for both iron and erbium confirms that the line locations are indeed close enough to randomly distributed within large bins. This also offers a rationale for the accuracy of the results done with the artificial line-broadening prescription of Fontes et al. [1, 2]: spreading line opacity over frequency is analogous to many narrow lines at random positions within a coarse bin, so either picture can yield similar cumulative opacity in the bin, consistent with smeared Sobolev tables reproducing Monte Carlo transport [3]. At low frequency resolution κ_R of the saturated expansion opacity and $\kappa_R^{\rm exp}$ are close because a single wide bin often sets the stopping distance. In that limit, Eq. (9) applies and the effective opacity is the same as the saturated opacity. Using the formalism formerly described, we calculated the opacity of materials for different times, densities and temperatures and constructed opacity tables that can be used for simulations of type Ia supernova and kilonova. To simulate Type Ia supernova ejecta, we created two tables for two compositions. The first for the outer layer of the supernova, which is a mixture of Silicon, Sulfur, Argon, and Calcium in the ratio of 0.535, 0.32, 0.062, and 0.083 respectively. The second for the core of the supernova, which is a decay of ^{56}Ni followed its decay chain, $^{56}\mathrm{Ni} \rightarrow {}^{56}\mathrm{Co} \rightarrow {}^{56}\mathrm{Fe}$, using the composition appropriate for each time window. We assumed the half-lives of $t_{1/2}(^{56}\mathrm{Ni}) = 6.075$ days and $t_{1/2}(^{56}\mathrm{Co}) = 77.233$ days. Kilonova ejecta contain essentially every r-process species, and computing a line list for each of them is impractical. We therefore built the kilonova table for a reduced mixture of eight representative elements derived directly from the solar r-process abundance pattern. We start from the solar r-process residuals $N_r(A)$, i.e. the solar isotopic abundances with the calculated s-process contribution subtracted, taken from Simmerer et al. [10] as tabulated by Sneden et al. [11]. Converting to mass, $M(A) \propto A\,N_r(A)$, we restrict the sum to $90 < A \leq 215$, which holds 20% of the mass of the tabulated pattern ($69 \leq A \leq 238$) and covers the second ($A \approx 130$) and third ($A \approx 195$) r-process peaks together with the lanthanides between them. This range is then divided into eight contiguous windows in A, each represented by the one element whose standard atomic weight falls inside it, and each representative is given the mass fraction accumulated in its own window. Figure 3 shows the cumulative mass as a function of A with the window boundaries and the resulting fractions. The eight representatives are Zr ($Z = 40$), Ru (44), Xe (54), Nd (60), Eu (63), Er (68), Re (75), and Pb (82), with mass fractions 0.049, 0.108, 0.460, 0.043, 0.030, 0.047, 0.206, and 0.057. Splitting by valence shell, the mixture is 51.7% p-block, 36.2% d-block, and 12.1% f-block, so the lanthanide mass fraction of the solar-derived mixture is $X_{\rm lan} = 0.121$. Because $X_{\rm lan}$ is the quantity that varies most strongly between kilonova ejecta components, we also built six rescaled tables in which the three lanthanides are scaled to a prescribed $X_{\rm lan}$ between 10^{-6} and 10^{-1}, keeping their relative proportions, and the five remaining elements are renormalized to $1 - X_{\rm lan}$ keeping theirs. These tables are used in Section 4.

Each table stores, at every (T, ρ, t), $\kappa_R^{\rm exp}$, $f_{\rm abs}$, $\kappa_P^{\rm abs}$, and $\kappa_P^{\rm tot}$. The fraction $f_{\rm abs}$ is the Rosseland mean of $\kappa_{\rm abs}/(\kappa_{\rm abs} + \kappa_{\rm scat})$ with $\kappa_{\rm abs} = \kappa_{\rm exp} - \kappa_{\rm T}$ and $\kappa_{\rm scat} = \kappa_{\rm T}$,

$$f_{\rm abs} = \frac{\int \frac{\kappa_{\rm abs}}{\kappa_{\rm abs}+\kappa_{\rm scat}} \frac{\partial B_\nu}{\partial T}\, d\nu}{\int \frac{\partial B_\nu}{\partial T}\, d\nu}. \tag{11}$$

In transport we use $\kappa_R^{\rm abs} = f_{\rm abs}\,\kappa_R^{\rm exp}$ and $\kappa_R^{\rm scat} = (1 - f_{\rm abs})\,\kappa_R^{\rm exp}$: absorption deposits energy and re-emits the packet (IMC), while scatter only redirects it. The last two entries are Planck means of the saturated opacity and are not used for the same purpose. $\kappa_P^{\rm abs}$ is the Planck mean of the saturated absorption opacity, $\kappa_{\rm sat} - \kappa_{\rm T}$; it sets the cell temperature, which is chosen so that the thermal emission $4\pi\kappa_P^{\rm abs}\rho B(T)$ balances the energy deposited in the cell. $\kappa_P^{\rm tot}$ is the Planck mean of the total saturated opacity, including Thomson scattering. It takes no part in the transport and is stored only so that the Planck and Rosseland weightings can be compared with the same physical processes included in both, as we do next.

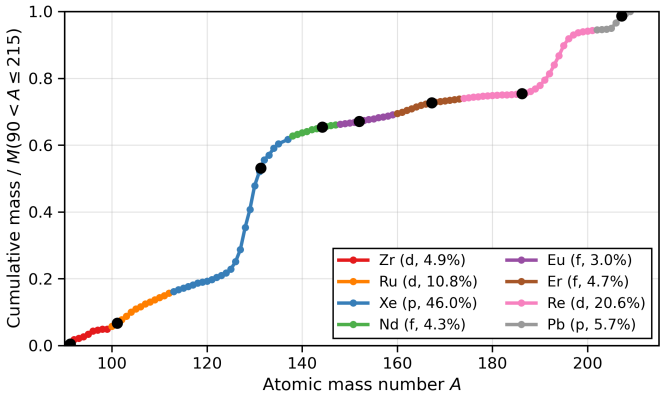


Figure 3: Cumulative mass of the solar r-process residual pattern [10, 11] as a function of mass number, normalized to the total mass in $90 < A \leq 215$. Colors mark the eight contiguous A windows and the legend gives the element chosen to represent each window, its valence block, and the mass fraction contained in the window. Black circles are placed at the standard atomic weight of each representative.

Since the ordinary Rosseland mean opacity is problematic to define for expansion opacity, previous works [4, 5, 6] often used the Planck mean opacity to compare the opacity of different compositions in different temperatures. We compare $\kappa_R^{\rm exp}$ to the Planck mean through their ratio as a function of temperature and density. For consistency in this comparison we use the Planck mean of the total saturated opacity, $\kappa_P^{\rm tot}$, including Thomson scattering, so that both means contain the same physical processes and the ratio tests how much the frequency weighting differs rather than whether scattering is included in one mean but not the other. Figure 4, Figure 5, and Figure 6 show $\kappa_R^{\rm exp}$ and the ratio $\kappa_P^{\rm tot}/\kappa_R^{\rm exp}$ for the ^{56}Ni decay chain, Type Ia outer-layer composition, and kilonova composition. At one day, the ratio spans approximately 0.05–39 for the Ni-rich core. For the kilonova composition it is 1.6–5 up to $T \simeq 0.8$ eV, but reaches 132 near $T \simeq 1.2$ eV at low density. Therefore for both cases the Planck mean opacity is not a good proxy for the Rosseland mean opacity.

The kilonova figures are cut at $T = 2$ eV because our atomic data contain only the first three ionization stages, so the mean

ionization is capped at $\langle Z\rangle = 3$. Above that temperature the material is held artificially at the third stage rather than being stripped further, and the opacity is incorrect. This does not affect the transport calculations of Section 4, because the emergent radiation is set by the diffusion surface and not by the optically thick interior: the plasma temperature at that diffusion surface never exceeds 1.8 eV in any of the runs (Figure 11), so the opacity that shapes the light curves is always taken from within the reliable range, even though the deeply buried cells are assigned higher temperatures. For the Type Ia outer-layer composition (Si, S, Ar, Ca) the ratio spans 0.96–5.2 over the plotted range. In this regime the opacity is dominated by Thomson scattering and free-free absorption, which depend only weakly on frequency, so the two averaging schemes give similar results. Figure 10 shows one-dimensional slices of $\kappa_R^{\rm exp}$ for the Type Ia outer-layer and kilonova compositions. For both compositions the opacity decreases with density. This follows from the structure of (9): each line contributes a blocking factor $1-\exp(-\tau_l)$ divided by ρct, and the Sobolev depth τ_l is itself proportional to ρt. A weak line with $\tau_l \ll 1$ therefore contributes in proportion to the number of absorbers, the two density factors cancel, and its opacity per unit mass is independent of ρ. A strong line with $\tau_l \gg 1$ instead saturates at $1-\exp(-\tau_l) \to 1$: it already blocks its entire frequency interval, and adding more absorbers cannot block any more. Once the strong lines dominate the sum, the total blocked fraction stops growing while the mass it is divided by keeps growing, so $\kappa_R^{\rm exp}$ falls roughly as $1/\rho$. The temperature dependence is not monotonic.

4. Monte Carlo Simulation With Single Group Opacity

We performed radiation transport simulations to test the effective expansion Rosseland opacity model. The simulations were performed using URILIGHT [12, 13, 14], a Monte Carlo radiative transfer code written by Elbaz et al. and designed for supernova ejecta. For this work we made several modifications to the code, including support for Rosseland and Planck mean opacity tables and related changes needed for the simulations presented here. With permission from the code writers, we published the modified version in a Git repository (`https://github.com/roee2000/Urilight_SN`). The Type Ia supernova model is one-dimensional and spherically symmetric and contains 100 cells in homologous expansion. The ejecta has two composition layers: an inner $0.98\ M_\odot$ layer of ^{56}Ni and an outer $0.4\ M_\odot$ layer of Si, S, Ar, and Ca. The simulation starts at $t = 2$ days and ends at $t = 80$ days with 100 logarithmic time steps.

Energy is deposited by the $^{56}\mathrm{Ni} \to {}^{56}\mathrm{Co} \to {}^{56}\mathrm{Fe}$ decay chain. Gamma-rays and positrons are transported at the beginning of the calculation, and their deposited energy is saved for each cell and time step. In the radiation transport stage, new optical photon packets are emitted according to the fraction of the total gamma-ray and positron energy deposited in each cell. Existing photon packets are propagated with effective scattering: when a packet is absorbed it is not removed, but is re-emitted with a frequency sampled according to the local cell emission spectrum.

The kilonova simulation uses a one-dimensional ejecta profile with total mass $0.05\ M_\odot$ and 32 cells, from $t = 0.05$ days to $t = 100$ days with 70 logarithmic time steps. The composition is uniform and is the eight-element solar-derived mixture described in Section 3. Instead of a radioactive decay chain, the heating is an external power law per unit mass [15], $\dot\epsilon = 10^{10}(t/1\ \mathrm{day})^{-1.3}\ \mathrm{erg\ s^{-1}\ g^{-1}}$. We repeated the kilonova simulation for the six rescaled compositions, with $X_{\rm lan}$ between 10^{-6} and 10^{-1}, using the matching opacity table in each case. The ejecta profile and the heating rate are identical in all six runs, so any difference in the light curve comes only from the opacity.

In the full frequency-group calculation, photon packets keep their frequency and interact with the frequency-dependent opacity groups. For that, at each time step the simulation calculates the ionization stages of each material and goes through all bound-bound lines to calculate the opacity for that temperature and density. In the single-group calculation, the photon mean free path is set by $\kappa_R^{\rm exp}$, interpolated from the opacity table using local (T, ρ, t). The table value $f_{\rm abs}$ splits this into absorbing and scattering Rosseland means ($\kappa_R^{\rm abs}$ and $\kappa_R^{\rm scat}$) as described above. The cell temperature is determined by requiring the emitted energy to match the deposited energy; for this step we use $\kappa_P^{\rm abs}$, the Planck mean of the saturated absorption opacity.

Both simulations are one-dimensional homologous expansions, so the spatial coordinate can be represented by the velocity coordinate $r = vt$ and the density scales as t^{-3}. The Type Ia model uses an exponential density profile matching the W7 model [16], $\rho(v,t) = \rho_{\rm Ia,0} t^{-3}\exp(-v/v_e)$, normalized to $1.38\ M_\odot$, with $v_e = 2.7\times10^8\ \mathrm{cm\ s^{-1}}$ $(0.009c)$ and an outer edge at $0.093c$. The kilonova model instead has a flat inner profile $\rho = \rho_0 t^{-3}$ for $v < 0.1c$ and $\rho = \rho_0 t^{-3}(v/0.1c)^{-4.5}$ for $0.1c < v < 0.4c$, with ρ_0 set so the total mass is $0.05\ M_\odot$. Thus the Type Ia ejecta are centrally concentrated and slower, while the kilonova has a flat core and a faster power-law envelope. The steep outer slope follows the ejecta-density consideration in [17].

Figure 7 compares the light curves obtained with the single-group calculation, using $\kappa_R^{\rm exp}$, to the full frequency-group calculation for both ejecta models. The one-group and multigroup results follow the same qualitative trends in both cases, showing that the averaged opacity captures the main transport behavior while avoiding the cost of recalculating the full frequency-dependent opacity during the transport calculation. For the kilonova we show three lanthanide fractions spanning four decades, $X_{\rm lan} = 10^{-5}$, 10^{-2}, and 10^{-1}. Raising $X_{\rm lan}$ suppresses the luminosity during the first few days by nearly an order of magnitude, as the additional line opacity traps the radiation, and the single-group calculation reproduces this trend over the whole range. The agreement is best for the lanthanide-rich cases, where the ejecta stay optically thick and the transport is closest to diffusion; the single-group maximum is 13% below the multigroup one for both $X_{\rm lan} = 10^{-2}$ and 10^{-1}. For $X_{\rm lan} = 10^{-5}$ the single-group light curve is 33% brighter, since with few lines the escaping radiation is dominated by the most transparent frequency windows, which a single transport

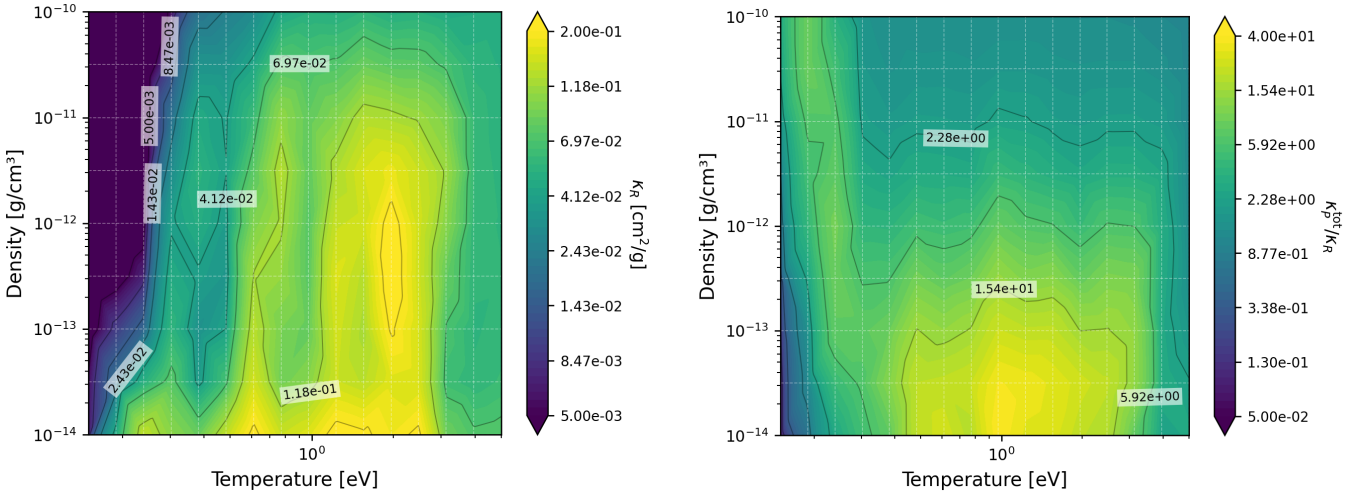


Figure 4: Type Ia core composition at $t = 1$ day: $\kappa_R^{\rm exp}$ (left) and $\kappa_P^{\rm tot}/\kappa_R^{\rm exp}$ ratio (right). Density 10^{-14}–10^{-10} g/cm^3.

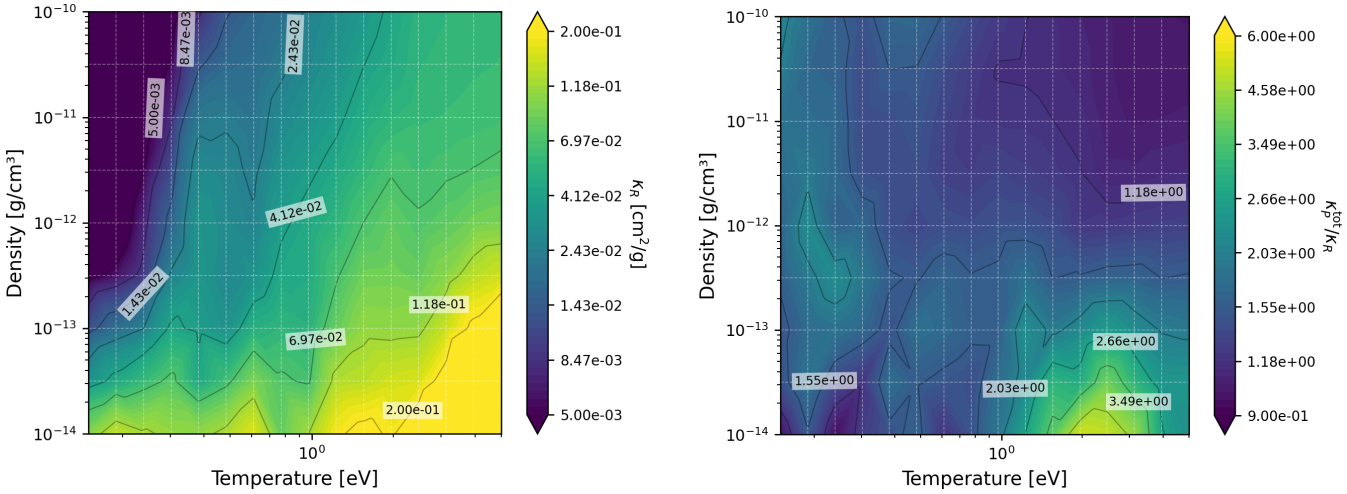


Figure 5: Type Ia outer-layer composition at $t = 1$ day: $\kappa_R^{\rm exp}$ (left) and $\kappa_P^{\rm tot}/\kappa_R^{\rm exp}$ ratio (right). The composition is Si, S, Ar, and Ca with mass fractions 0.535, 0.32, 0.062, and 0.083. Density 10^{-14}–10^{-10} g/cm^3.

opacity cannot represent. After about 10 days all curves converge, because the ejecta become optically thin and the luminosity simply tracks the heating rate regardless of the opacity.

Figure 8 compares the radiation temperature measured from escaping packets in the multigroup Type Ia calculation to the plasma temperature at the effective photospheric surface in the single-group calculation. The single-group calculation recovers the multigroup $T_{\rm rad}$ well. We show two choices for the effective path factor, $f = 1$ and $f = 3$, to bracket the geometrical uncertainty. At early times the escaping radiation is strongly affected by the outer ejecta layers, where scattering dominates absorption and $\tau_{\rm abs} \ll \tau_{\rm scat}$. Photons therefore undergo many scatterings and the escape problem is closer to the three-dimensional diffusion limit. At late times, when $\tau_{\rm abs} \sim 1$, the last absorption surface is closer to the one-dimensional radial limit.

Figure 9 maps $T_{\rm plasma}$ and $\kappa_R^{\rm exp}$ in the Type Ia validation simulation on the (v, t) grid. The temperature is taken from the URILIGHT profiles. The opacity in each cell is interpolated from the precomputed NiCoFe and SN1A tables using local (T, ρ, t), with ρ from homologous expansion and the ejecta geometry. The Ni/IME boundary is marked by a black dashed line. With time, the ejecta cools and the outer layers become more transparent.

On the opacity panel, the red curves are the $\tau = 1$ surfaces: at each t we integrate inward from the outer edge until $\int \kappa_R^{\rm exp}\, \rho\, f\, dR = 1$, where each velocity shell has radial thickness $\Delta R_i = t\, \Delta v_i$. We show $f = 1$ and $f = 3$ to bracket the radial and three-dimensional diffusion limits. The factor $f = 3$ comes from the diffusion coefficient, $D = c/(3\kappa\rho)$: equivalently, diffusion in three dimensions behaves like a radial transport problem with an effective opacity larger by a factor of three.

The cyan curves mark where the random-walk photon leakage time equals the dynamical time, $t_{\rm phot} = t$, for the same two choices of f. Using the same inward integration, the cumulative optical depth to shell i is $\tau_i = \tau_{i+1} + f\kappa_{R,i}^{\rm exp}\, \rho_i\, \Delta R_i$, with

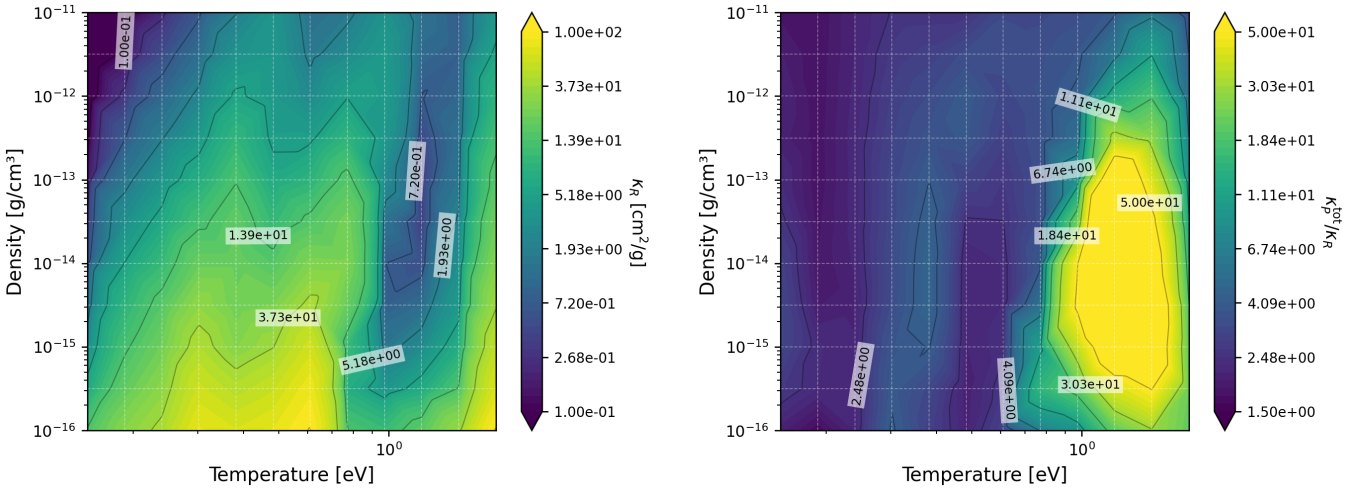


Figure 6: Solar-derived kilonova composition ($X_{\rm lan} = 0.121$) at $t = 1$ day: $\kappa_R^{\rm exp}$ (left) and $\kappa_P^{\rm tot}/\kappa_R^{\rm exp}$ ratio (right). Density 10^{-16}–10^{-11} g/cm^3, the range spanned by the diffusion surface in the transport calculations of Section 4. The temperature range is cut at 2 eV, above which the mean ionization saturates at the three-electron limit of our atomic data (see text). Both colorbars are clipped at both ends; within the plotted range $\kappa_R^{\rm exp}$ spans 0.006–103 cm^2/g and the ratio spans 1.6–132.

$\tau = 0$ at the outer edge. A diffusing photon does not cross a shell once: each scatter can send it back, so on average it traverses the thickness $(2\tau_i + 1)$ times before leaking outward. Each traversal takes time $\Delta R_i/c$, so

$$\Delta t_i = \frac{(2\tau_i + 1)\,\Delta R_i}{c}. \tag{12}$$

Starting from the outer edge and accumulating inward, we locate the velocity where $t_{\rm phot} = t$. Together, the two curves mark the last-scattering surface and the diffusion depth relevant at time t.

4.1. Runtime Advantage of Single-Group Transport

The single-group calculation is faster for three main reasons. First, each photon packet samples a Rosseland-mean transport opacity, so fewer packets are needed to obtain a stable estimate of the frequency-averaged energy transport. In a multigroup calculation, the same average behavior is recovered only statistically from packets distributed over frequency, which requires more packets to reduce numerical noise. Second, each packet propagation is cheaper. In a line-rich multigroup calculation, photons can repeatedly redshift into dense groups of bound-bound lines, be absorbed after very short path lengths, and be re-emitted until they enter a more transparent frequency window. These repeated short propagation steps add many microsteps inside a single time step. The single-group calculation replaces this behavior with an averaged mean free path, so packet histories contain fewer, more uniform transport steps. Third, the multigroup calculation must recompute the frequency-dependent opacity for every cell from its local (T, ρ, t) by going through the bound-bound line list of the material. The single-group calculation instead interpolates one value from a precomputed opacity table.

The speedup depends on which part of the multigroup calculation is the bottleneck. For lanthanide-rich kilonova ejecta, where photon transport through many bound-bound lines can dominate the runtime, the single-group calculation can be about 30 times faster and has significantly smaller numerical noise for the same number of packets. In cases where the main multigroup cost is recalculating the opacity itself, the gain can be even larger; for example, the Type Ia calculation shown in Figure 7 was about 100 times faster in the single-group calculation. This advantage is strongest when the same composition is used many times, since the opacity table can be reused. If many different compositions must be explored, a separate table has to be built for each composition, and this preprocessing can itself require significant runtime.

5. Typical Effective Rosseland Opacity

Gray kilonova models commonly replace the frequency-dependent opacity by one constant assigned to each ejecta component. Values used in the literature span approximately 0.2–30 cm^2/g [3]; a frequently used distinction is $\kappa \simeq 0.5$–1 cm^2/g for lanthanide-poor (blue) material and $\kappa \simeq 10$ cm^2/g for lanthanide-rich (red) material [18, 19, 20]. Such values are useful in analytic and semi-analytic light-curve models, but they are generally chosen by composition class or calibrated against a light curve. For example, Hotokezaka & Nakar [17] use $\kappa = 0.5$ cm^2/g outside $0.2c$ and 3 cm^2/g inside it, while Just et al. [5] prescribe a broken power law in $X_{\rm lan}$ calibrated to bolometric light curves. Detailed transport calculations also show that the gray value is not universal: Banerjee et al. [21] find approximately 0.8 cm^2/g for lanthanide-free material and 1–10 cm^2/g for lanthanide-rich material. Our method instead determines the gray opacity self-consistently. At each time it locates the diffusion surface, evaluates its temperature and density, and uses $\kappa_R^{\rm exp}(T, \rho, t)$ to calculate the effective opacity of the leaking envelope.

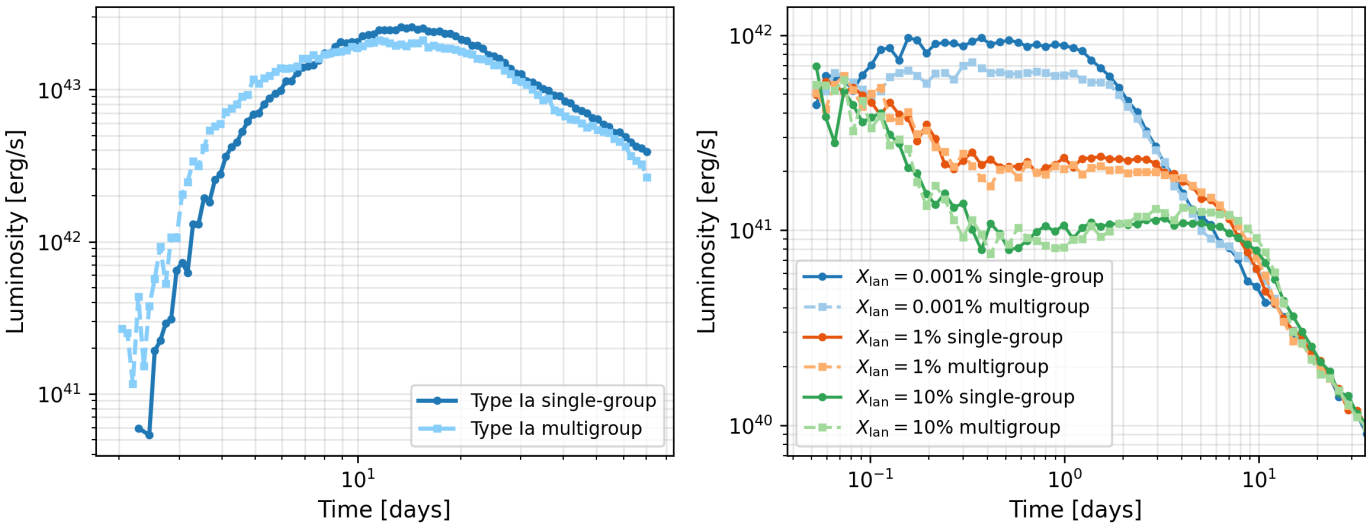


Figure 7: Bolometric luminosity from the Type Ia (left) and kilonova (right) validation simulations. For each model we compare the single-group calculation using $\kappa_R^{\rm exp}$ (solid, circles) with the corresponding full frequency-group calculation (dashed, squares). The kilonova runs share the same ejecta profile and heating rate and differ only in the lanthanide mass fraction of the composition, and hence in the opacity table. The final observer times, which can receive photons emitted after the simulation ends, are excluded by requiring $t < t_{\rm end}(1-\beta_{\rm max})$.

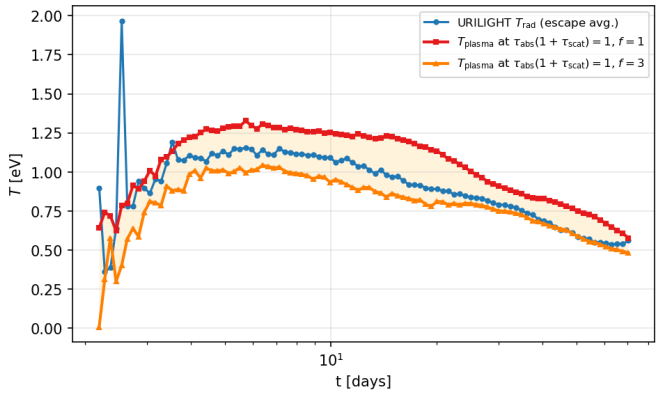


Figure 8: Radiation temperature in the Type Ia validation simulation. The blue curve shows the multigroup URILIGHT escape-averaged $T_{\rm rad}$. The red and orange curves show the single-group plasma temperature at $\tau_{\rm abs}(1+\tau_{\rm scat}) = 1$, using path factors $f = 1$ and $f = 3$.

5.1. *Why an approximately constant opacity produces a power-law light curve*

The nearly flat kilonova light curves in the right panel of Figure 7 can be understood analytically. Consider homologous ejecta with constant gray opacity κ, density $\rho(v,t) = At^{-3}v^{\alpha}$, and heating per unit mass $\dot{\epsilon} \propto t^{-\beta}$. The optical depth exterior to velocity v is

$$\tau(v,t) = \int_v^{v_{\rm max}} \kappa\rho\, t\, dv' \propto \kappa t^{-2} v^{\alpha+1}, \tag{13}$$

where the outer-boundary term is negligible for $\alpha < -1$ and $v \ll v_{\rm max}$. With $t_{\rm diff} \sim \tau v t/c$, the diffusion surface $t_{\rm diff} = t$ follows

$$v_{\rm diff}(t) \propto \left(\frac{t^2}{\kappa}\right)^{1/(\alpha+2)}. \tag{14}$$

The mass exterior to this surface is $M(>v_{\rm diff}) \propto v_{\rm diff}^{\alpha+3}$ for $\alpha < -3$. While the diffusion surface is inside the power-law envelope, the luminosity is therefore

$$L(t) \simeq \dot{\epsilon}(t) M(>v_{\rm diff}) \propto \kappa^{-(\alpha+3)/(\alpha+2)} t^{-\beta+2(\alpha+3)/(\alpha+2)}. \tag{15}$$

For our $\alpha = -4.5$ profile and $\beta = 1.3$ heating law, $L \propto \kappa^{-0.6} t^{-0.1}$, so the luminosity is almost constant while the diffusion surface moves through the power-law envelope. This is the same physical scaling used in analytic kilonova models [22]. It explains both features of the kilonova panel in Figure 7: each composition produces an extended luminosity plateau, while increasing $X_{\rm lan}$ raises κ and lowers the plateau approximately as $\kappa^{-0.6}$. Once the ejecta become transparent, all of the deposited energy escapes and L instead approaches $M_{\rm ej}\dot{\epsilon} \propto t^{-1.3}$, as seen after approximately ten days in Figure 7.

5.2. *Opacity sampled by the diffusion surface*

At each time we define the effective gray value as the column-mass-weighted Rosseland opacity outside the surface $t_{\rm diff} = t$,

$$\kappa_{\rm avg}(t) = \frac{\displaystyle\int_{v_{\rm diff}}^{v_{\rm max}} \kappa_R^{\rm exp}\rho\, t\, dv}{\displaystyle\int_{v_{\rm diff}}^{v_{\rm max}} \rho\, t\, dv} = \frac{\tau(v_{\rm diff})}{\Sigma(v_{\rm diff})}. \tag{16}$$

This is the constant opacity that reproduces the same optical depth through the leaking envelope at that time.

Figure 11 shows the velocity, temperature, and density sampled by the kilonova diffusion surface. Increasing $X_{\rm lan}$ keeps the surface at larger velocity and lower density for longer. At the earliest times all six surfaces lie in the high-temperature regime, $T_{\rm eff} \simeq 1.5$–1.8 eV, where the opacity tables for the

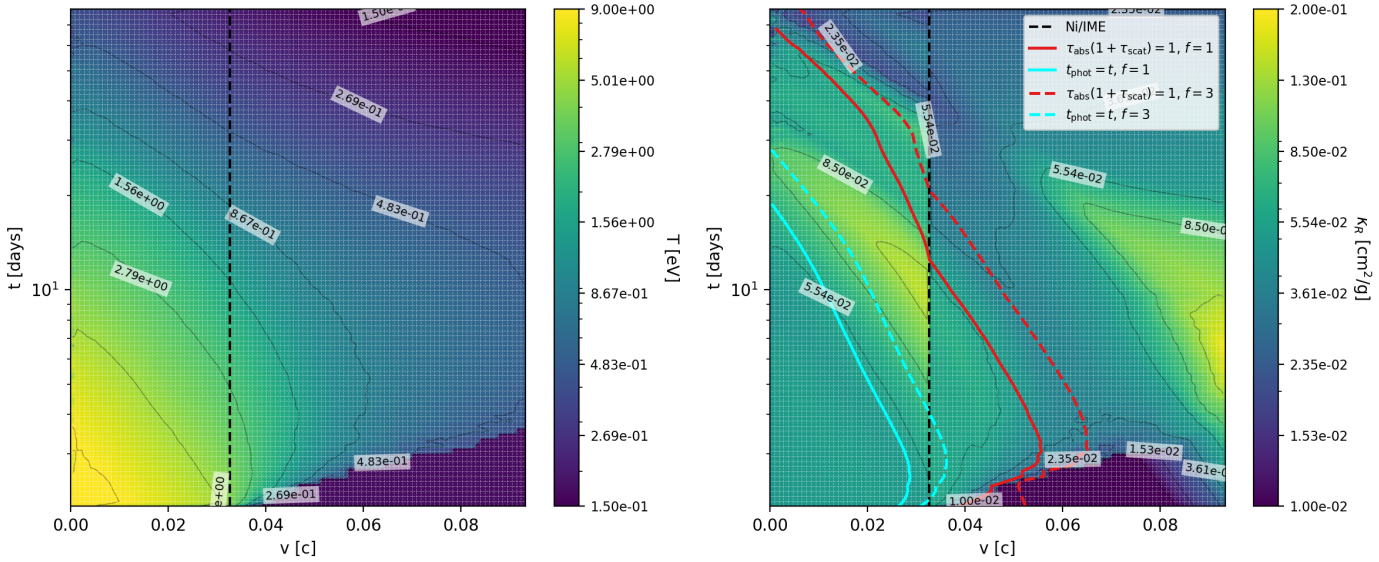


Figure 9: $T_{\rm plasma}$ (left) and $\kappa_R^{\rm exp}$ (right) in the Type Ia validation simulation. The black dashed line marks the Ni/IME boundary. On the opacity map, red curves show the $\tau_{\rm abs}(1+\tau_{\rm scat})=1$ surface and cyan curves show the $t_{\rm phot}=t$ surface. Solid curves use $f=1$ and dashed curves use $f=3$ (see text).

different $X_{\rm lan}$ converge. The early $\kappa_{\rm avg}$ values consequently differ by less than a factor of two, and the light curves in Figure 7 initially show little lanthanide dependence. As the surface cools, the lanthanide contribution becomes important and the luminosities separate. We emphasize that $T_{\rm eff}$ does not exceed 1.8 eV; hotter cells are deeper than the diffusion surface and do not set the emerging luminosity.

Figure 10 isolates the temperature and density dependence. At fixed $\rho = 10^{-13}$ g/cm^3, increasing $X_{\rm lan}$ raises the opacity strongly below $T \simeq 1$ eV. Around $T \simeq 1.2$ eV, however, all five kilonova curves fall toward the same opacity: they differ by less than a factor of two despite five decades in $X_{\rm lan}$. At this ionization transition the additional lanthanide lines no longer fill the Rosseland-weighted frequency windows, and the non-lanthanide background sets the opacity floor. This convergence explains why the earliest diffusion-surface opacity and luminosity depend only weakly on $X_{\rm lan}$. The fixed-temperature slice shows that away from this transition the composition ordering remains over the full density range.

Figure 12 shows $\kappa_{\rm avg}(t)$. For the Type Ia calculation the diffusion surface is already inside the ^{56}Ni-rich layer at the first simulated epoch: $v_{\rm diff} < 0.029c$, whereas the Ni/outer-layer boundary is at $0.033c$. We therefore quote only the local Ni-rich opacity at that surface, $\kappa_R^{\rm exp}(v_{\rm diff}) \simeq 0.05$–$0.07$ cm^2/g over the simulated interval. For the kilonova, after the short high-temperature phase, each composition has an extended interval over which $\kappa_{\rm avg}$ changes slowly. The representative values averaged over the interval $0.1c < v_{\rm diff} < 0.3c$ are 0.36, 0.37, 0.39, 0.99, 4.64, and 18.45 cm^2/g for $X_{\rm lan} = 10^{-6}$, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, and 10^{-1}, respectively. These values naturally recover the usual lanthanide-poor and lanthanide-rich opacity scales without calibrating to the light curves.

5.3. *A composition-dependent gray-opacity prescription*

To test whether the inferred values depend strongly on the assumed ejecta profile, we repeated the six-composition calculation for outer density slopes $\alpha = -3.5$, -4.5, and -5.5, and for ejecta masses 0.1 and 10 times the fiducial mass. For each run we average (16) over the time during which $0.1c < v_{\rm diff} < 0.3c$, weighting uniformly in $\ln t$. This selects the power-law-envelope phase described by (15) and excludes the final transparent phase, when the luminosity no longer depends on opacity.

A single fit in $\ln\kappa$ to all 30 simulations gives

$$\boxed{\kappa_{\rm avg}(X_{\rm lan}) = 0.34 + 99\,X_{\rm lan}^{0.72}\ {\rm cm^2\,g^{-1}}.} \qquad (17)$$

Figure 13 shows that composition produces a nearly two-order-of-magnitude change, whereas changing the density profile mainly shifts the normalization: the rms scatter around (17) is 32%, and the largest deviation is a factor of 1.8. Thus the functional dependence on $X_{\rm lan}$ is robust to these spherical ejecta variations, although the formula should not be interpreted as an exact universal opacity. Its sublinear exponent and low-$X_{\rm lan}$ floor are consistent with the saturation of strong lanthanide lines discussed by Kasen et al. [23] and with the empirical prescription of Just et al. [5]; the difference is that (17) is derived from the Rosseland transport opacity rather than fitted to a bolometric light curve.

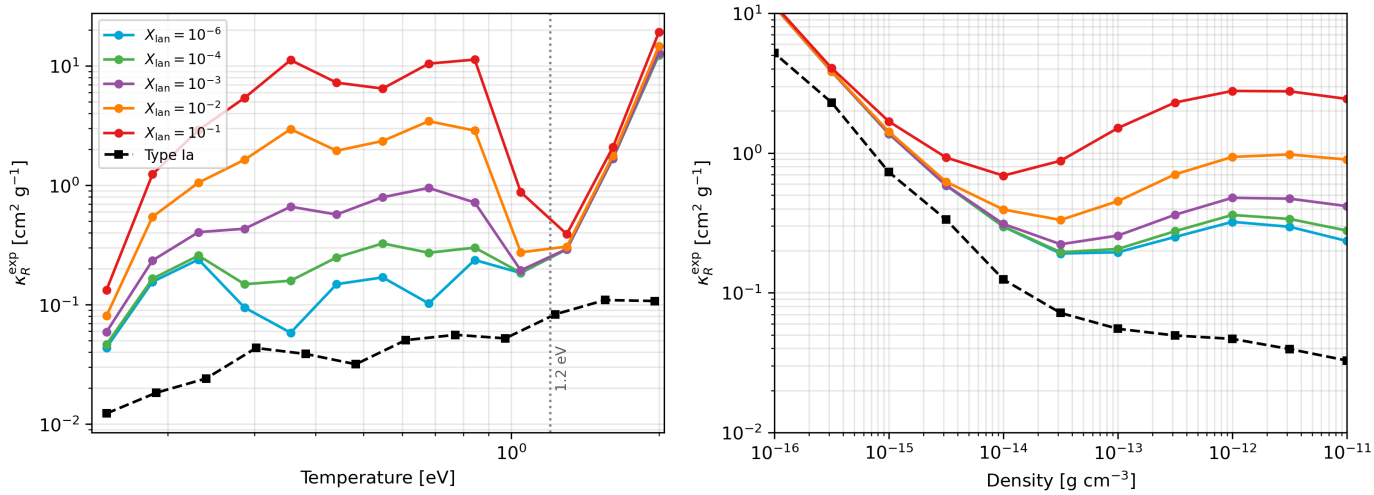


Figure 10: Effective expansion Rosseland opacity for the five kilonova compositions and one Type Ia outer-layer composition. Left: temperature dependence at $\rho = 10^{-13}$ g/cm^3; the dotted line marks $T = 1.2$ eV. Right: density dependence at $T = 1$ eV. Both compositions are evaluated at exactly one day.

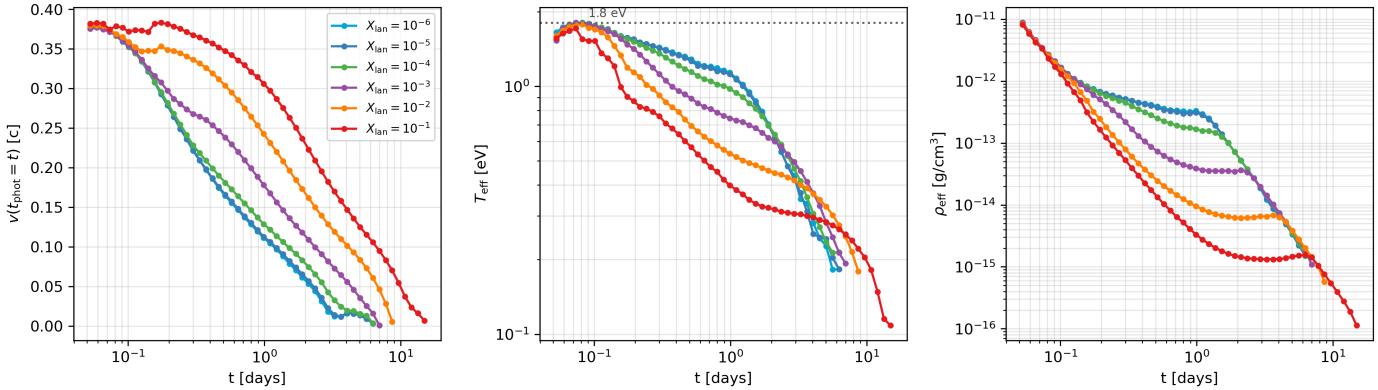


Figure 11: Velocity (left), plasma temperature (middle), and density (right) at the kilonova diffusion surface $t_{\rm diff} = t$, for six lanthanide mass fractions. The dotted line marks $T = 1.8$ eV, the maximum temperature reached by the diffusion surface.

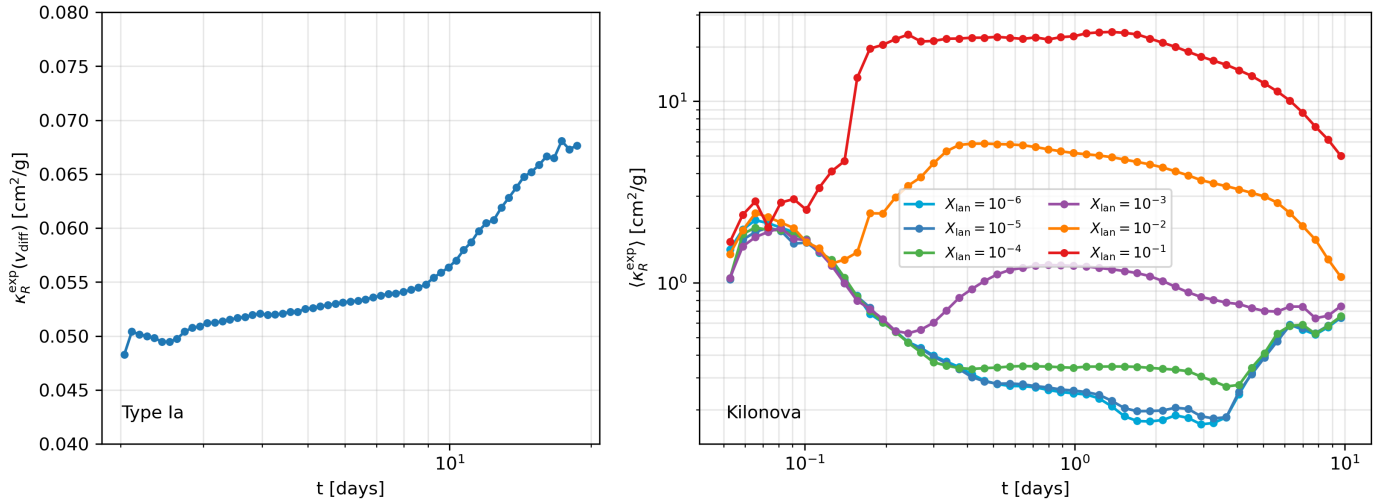


Figure 12: Effective expansion Rosseland opacity sampled by the diffusion region. Left: local opacity at the Type Ia diffusion surface, which remains inside the Ni-rich layer throughout the calculation. Right: column-averaged opacity outside the kilonova diffusion surface for the six compositions.

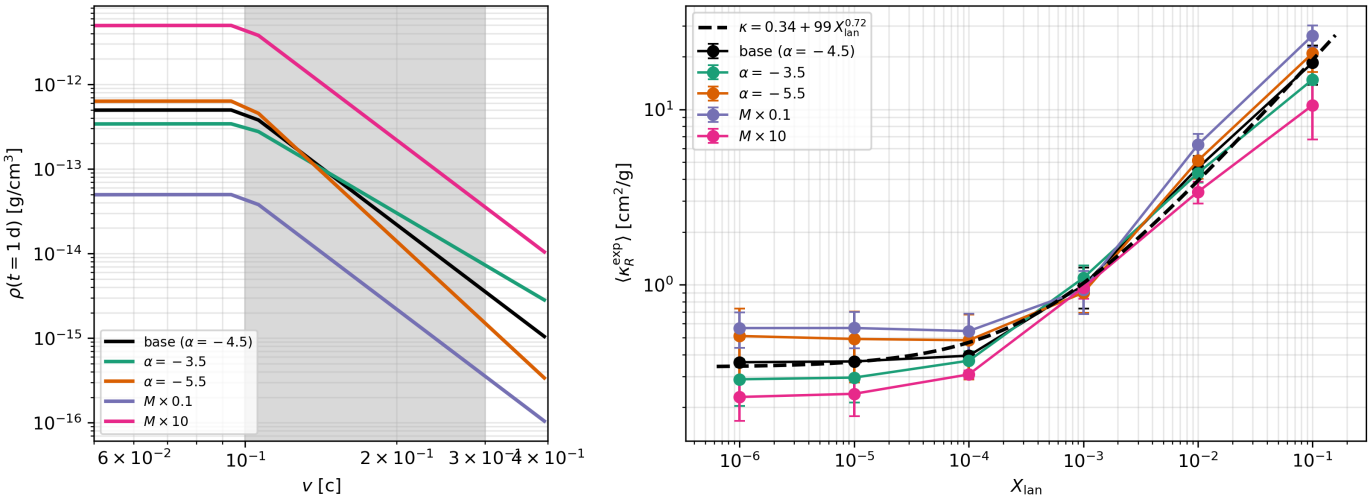


Figure 13: Robustness of the composition-dependent gray opacity. Left: density profiles at one day; the shaded region marks $0.1c < v < 0.3c$. Right: logarithmic-time-weighted mean and standard deviation of κ_{avg} while the diffusion surface crosses that interval. Colors identify ejecta profiles rather than compositions. The dashed curve is one fit to all 30 simulations, given by (17).

6. Summary

We defined the effective expansion opacity, $\kappa_{\rm exp}$, for media in homologous expansion. The opacity is defined from the average distance an outward travelling photon needs to reach optical depth unity, $\kappa_{\rm exp} = 1/(\rho \Delta x)$. Taking the Rosseland mean of $\kappa_{\rm exp}$ gives $\kappa_R^{\rm exp}$, which is not dependent on the frequency resolution. We showed this for both iron and erbium, representing iron-group Type Ia material and lanthanide-rich kilonova material. We also compared $\kappa_R^{\rm exp}$ with the Planck mean opacity for Type Ia core, Type Ia outer-layer, and kilonova compositions. The ratio between the two depends strongly on temperature and density, showing that the Planck mean is not a reliable replacement for $\kappa_R^{\rm exp}$. The disagreement is largest for the kilonova composition, where the ratio spans more than three orders of magnitude across the tabulated range.

For the kilonova we replaced the usual ad hoc element mixture with a composition of eight representative elements derived from the solar r-process residual abundance pattern, and built a family of tables in which only the lanthanide mass fraction is varied, from $X_{\rm lan} = 10^{-6}$ to 10^{-1}.

Following the opacity sampled by the diffusion surface, we found representative kilonova values from 0.36 cm^2/g at $X_{\rm lan} = 10^{-6}$ to 18.45 cm^2/g at $X_{\rm lan} = 10^{-1}$. Varying the ejecta mass and outer density slope gives the practical prescription $\kappa_{\rm avg} = 0.34 + 99 X_{\rm lan}^{0.72}$ cm^2/g with 32% rms scatter over the tested spherical profiles.

Finally, we tested single-group radiation transport simulations using the new opacity tables in URILIGHT. The single-group light curves reproduce the main behavior of full frequency-group calculations for both Type Ia supernova and kilonova ejecta, and for the kilonova they follow the change in the light curve across the full range of lanthanide fractions.

The Python code for building opacity tables using this algorithm is available at https://github.com/roee2000/expansion_opacity_calculation.

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